

Requirements							
Req ID	Requirement Title	Revision	Change Classification	Source Baseline	Change Rationale	Requirement Intent	Version
ACC-REQ-001	System Initialization	Rev. 01	NI	Ver. 01	NI	On system start, the controller shall enter the Power_Off with compressor_on = 0, fan_on = 0, fault_indicator = 0, and status outputs reset.	Ver. 01
ACC-REQ-002	Temperature Setpoint Input	Rev. 01	NI	Ver. 01	NI	The controller shall use set_temperature as the desired room temperature reference for cooling control.	
ACC-REQ-003	Power ON Command	Rev. 02	Applicability clarification	Ver. 01	Power_On now also contains standby and Fan-only operation.	When power_button == 1, the controller shall enter Power_On, set fault_indicator = 0, update set_temperature from desired_room_temperature, and compute temperature_error = room_temperature - set_temperature. The controller shall evaluate cooling demand only while cooling control is enabled.	
ACC-REQ-004	Power OFF Command	Rev. 01	NI	Ver. 01	NI	When power_button == 0, the controller shall enter the Power_Off state and set compressor_on = 0 and fan_on = 0.	
ACC-REQ-005	Cooling Demand Activation	Rev. 02	Applicability clarification	Ver. 01	Cooling-demand hysteresis now applies only while cooling control is enabled.	While cooling control is enabled, if temperature_error >= temperature_hysteresis_band, the controller shall set cooling_demand = 1.	
ACC-REQ-006	Cooling Demand Deactivation	Rev. 02	Applicability clarification	Ver. 01	Cooling-demand hysteresis now applies only while cooling control is enabled.	While cooling control is enabled, if temperature_error <= -temperature_hysteresis_band, the controller shall set cooling_demand = 0.	
ACC-REQ-007	Compressor Minimum OFF Time	Rev. 01	NI	Ver. 01	NI	After the compressor turns OFF, the controller shall prevent compressor restart until compressor_min_off_time has elapsed.	
ACC-REQ-008	Compressor Activation	Rev. 01	NI	Ver. 01	NI	If cooling_demand == 1 and the compressor is not in minimum OFF-time lockout, the controller shall set compressor_on = 1.	
ACC-REQ-009	Compressor Deactivation	Rev. 01	NI	Ver. 01	NI	If cooling_demand = 0, power_button == 0, or a fault condition is active, the controller shall set compressor_on = 0.	
ACC-REQ-010	Fan Activation During Compressor Activation	Rev. 01	NI	Ver. 01	NI	When compressor_on == 1, the controller shall set fan_on = 1.	
ACC-REQ-011	Fan Behavior After Compressor Deactivation	Rev. 01	NI	Ver. 01	NI	When compressor_on == 0 after active cooling, the controller shall keep fan_on = 1 for configured fan_run_on_time, then set fan_on = 0.	
ACC-REQ-012	Default Power-On Standby Behavior	Rev. 01	New Ver. 02 requirement	Ver. 02	NI	When power_button == 1, no fault condition is active, and mode_enabled == 0, the controller shall activate Power_On_Standby, set cooling_demand = 0, compressor_on = 0, compressor_lockout = 0, fan_on = 1, and fan_speed_cmd = 1 corresponding to Low Speed. The controller shall set fan_only_mode_enabled = 0, cooling_mode_enabled = 0, and automatic_mode_enabled = 0.	Ver. 02
ACC-REQ-013	Fan-Only Mode Selection and Response	Rev. 01	New Ver. 02 requirement	Ver. 02	NI	When power_button == 1, mode_enabled == 1, and mode_selected == 0, the controller shall activate Fan_Only_Mode, set fan_only_mode_enabled = 1, set cooling_mode_enabled = 0 and automatic_mode_enabled = 0, disable cooling control, set cooling_demand = 0, compressor_on = 0, compressor_lockout = 0, and keep fan_on = 1. The controller shall determine fan_speed_cmd from fan_speed_user_input.	
ACC-REQ-014	Cooling Mode Selection and Response	Rev. 01	New Ver. 02 requirement	Ver. 02	NI	When power_button == 1, mode_enabled == 1, and mode_selected == 1, the controller shall activate Cooling_Mode, set cooling_mode_enabled = 1, set fan_only_mode_enabled = 0 and automatic_mode_enabled = 0, and enable the cooling control mechanism. The controller shall determine cooling_demand using temperature_error and temperature_hysteresis_band, control the compressor using cooling demand and minimum-OFF-time protection, control the fan using compressor operation and fan run-on behavior, and determine fan_speed_cmd from fan_speed_user_input.	
ACC-REQ-015	Automatic Mode Selection and Response	Rev. 01	New Ver. 02 requirement	Ver. 02	NI	When power_button == 1, mode_enabled == 1, and mode_selected == 2, the controller shall activate Automatic_Mode, set automatic_mode_enabled = 1, set fan_only_mode_enabled = 0 and cooling_mode_enabled = 0, and enable the cooling control mechanism. The controller shall determine cooling demand automatically using the configured temperature logic. The controller shall select fan speed automatically using manual_fan_override == 0 and shall use manual fan-speed selection when manual_fan_override == 1.	
ACC-REQ-016	Automatic Fan Speed Selection	Rev. 01	New Ver. 02 requirement	Ver. 02	NI	While Automatic_Mode is active and manual_fan_override == 0, the controller shall set automatic_fan_speed_enabled = 1 and determine fan_speed_cmd as follows: If fan_on == 0, set fan_speed_cmd = 0. If fan_on == 1 and compressor_on == 0, set fan_speed_cmd = 1. If compressor_on == 1 and temperature_error < medium_speed_threshold, set fan_speed_cmd = 1. If compressor_on == 1, temperature_error >= medium_speed_threshold, and temperature_error < high_speed_threshold, set fan_speed_cmd = 2. If compressor_on == 1 and temperature_error >= high_speed_threshold, set fan_speed_cmd = 3.	
ACC-REQ-017	Manual Fan Speed Selection and Override	Rev. 01	New Ver. 02 requirement	Ver. 02	NI	While Fan_Only_Mode or Cooling_Mode is active, or while Automatic_Mode is active with manual_fan_override == 1, the controller shall determine fan_speed_cmd from fan_speed_user_input. If fan_on == 0, set fan_speed_cmd = 0. If fan_speed_user_input == 0, set fan_speed_cmd = 1. If fan_speed_user_input == 1, set fan_speed_cmd = 2. If fan_speed_user_input == 2, set fan_speed_cmd = 3. For any unsupported fan_speed_user_input, set fan_speed_cmd = 1 as the default fallback.	
ACC-REQ-018	Compressor Lockout Status	Rev. 01	New Ver. 02 requirement	Ver. 02	NI	While the compressor minimum-OFF-time lockout is active, the controller shall set compressor_lockout = 1 and keep compressor_on = 0. After compressor_min_off_time has elapsed, the controller shall set compressor_lockout = 0. The controller shall also set compressor_lockout = 0 when cooling control is disabled, during Power_Off, and during Fault.	Ver. 02
ACC-REQ-019	Fresh Cooling-Control Activation Lockout	Rev. 01	New Ver. 02 requirement	Ver. 02	NI	When cooling control transitions from disabled to enabled, the compressor mechanism shall start in minimum-OFF-time lockout with compressor_on = 0 and compressor_lockout = 1. Compressor activation shall not be permitted until compressor_min_off_time has elapsed and cooling demand is active.	